

Generative AI and DevOps Engineer

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Professional SUMMARY:

Highly skilled and results-driven **Generative AI and DevOps Engineer** with 8 years of progressive experience designing, implementing, and optimizing scalable AI solutions and DevOps processes. Proficient in integrating cutting-edge generative AI models into business workflows to drive innovation and efficiency. Demonstrates expertise in CI/CD pipelines, cloud platforms, infrastructure automation, and AI model deployment. Adept at collaborating with cross-functional teams to deliver high-impact solutions while ensuring reliability, security, and scalability. I am known for leveraging advanced tools and methodologies to enhance system performance and operational excellence.

TECHNICAL SKILLS:

Artificial Intelligence and Machine Learning: Data cleaning, Preprocessing, Data Manipulation, Linear Regression, Logistic Regression, Decision Trees, Random Forest, Bagging and Boosting, Deep learning, Computer Vision- CNN, Clustering, Natural Language Processing, Azure Machine Learning Studio, Azure AutoML, Docker containerization and Kubernetes, Python, ML Ops SQL, Explainable AI, SHAPLEY, tree explainer, XGBoost Regressor, Azure Cognitive Services, LangChain, Computer Vision, Custom vision, OCR, Text Analytics, Content Moderator, Image and Video Processing, Azure LUIS, Semantic Analysis, REST API, Bot Framework, Bot Emulator, ML Ops CI/CD.

Statistical Method: Predictive Analysis, Hypothesis Testing, Principal Component Analysis, Dimensionality Reduction, Market basket Analysis

Programming Language: Python - Numpy, Pandas, Scikit, Matplotlib, Seaborn, Tensorflow, Keras, Artificial Neural Network, Convolutional Neural Network, PyTorch, LSTM, SQL, Generative AI-ChatGPT, Prompt Engineering.

Software: Jupiter Notebook, Google Collaboratory, PyCharm, Spark, Tableau, R, Minitab, MS Excel, MS Powerpoint, SSMS - SQL, MS SQL, GCP, Vertex AI, Python, PyTorch, Kubernetes, Docker, Power BI, Power Big Query Editor, Azure Machine Learning Studio, Azure Designer, Azure auto ML, Git, GitHub

Cloud Platforms : Azure OpenAI, Amazon SageMaker, AWS BedRock, AWS Texttract, AWS Comprehend, S3, Google Cloud & IBM Watson Studio

Framework/DevOps tools: Maven, JIRA, SonarQube, Nexus, Jenkins, VSTS, Slack, TeamCity.

Libraries & Packages: SpaCy, Hugging Face, Sklearn, NLTK, ,Flask, NumPy, Pandas, Matplotlib , Seaborn.

Build Tools: Ant, Maven, Gradle, MSBuild

SCM: Git, GitHub, Bitbucket, GitLab

IAC Tools: Puppet, Ansible, Terraform

Gen AI Engineer: LLM APIs, Foundational Models, Generative AI Chatbots, NLQ Applications,

Application/Web Servers: Tomcat, WebLogic 9.x/10.x/12c, Apache 2.x/1.3.x, JBoss 7.1

Operating Systems: HP-UX, Solaris 10, Ubuntu 16.0.4, Red Hat Linux, Windows, Windows Server,

Database: Oracle, SQL Server, MySQL, Postgres

Education Details:

Master's in computer science

PROFESSIONAL EXPERIENCE:

Client: Bank Of America, Charlotte, NC

Aug 2022 to Till Date

Role: Gen AI Engineer

Responsibilities:

- Developed and performed end-to-end logic, design, and implementation for PoC of Inventory purchase forecast to reduce food wastage using Machine Learning solution on a cloud platform.
- Communicate with stakeholders' business VPs, Product managers, and Clients to understand business challenges that can be resolved with ML-based solutions.
- Retrieved/fetched data from multiple databases, understand the relations between the different entities, and stored procedures. Extract data using web scraping and integrate it with a dataset.
- Performed data analysis, and data visualization to understand the trends and patterns, of anomaly in the data.
- Designed, built, and trained ML- Artificial intelligence models such as Linear Regression, Decision Tree, Random Forest Regressor, Boosted Decision Trees regressor, LightGBM, XGBoost Regressor, Time series forecasting -ARIMA, ARIMAX. Performed hyperparameter tuning using Grid Search and Randomized Search cross-validations to optimize model performance.
- Working with AI clients for utilizing Azure cognitive microservices for Computer Vision use cases like image tag/ analytics, content moderation, Text extraction (OCR), and face recognition.
- Working with Azure LUIS APIs for language detection, phrase extraction, entity recognition.
- Proficient in Azure Machine Learning Studio for building, training, and deploying machine learning models.
- Experienced in using Azure SDK for Python to integrate machine learning workflows into cloudbased applications.
- Strong understanding of machine learning concepts, including supervised and unsupervised learning techniques.
- Expertise in random forest algorithms for classification and regression tasks.
- Hands-on experience with Python libraries such as scikitlearn for implementing machine learning models.
- Proficient in using pandas for data manipulation, cleaning, and preprocessing.
- Skilled in numpy for numerical computations and handling large datasets efficiently.

- Experienced in matplotlib for creating data visualizations and plots.
- Familiar with audio data processing using the librosa library for feature extraction and analysis.
- Knowledgeable in PyDub for audio file manipulation and processing.
- Capable of designing end-to-end machine learning pipelines using Python and Azure tools.
- Working with different data structures. Experienced in designing and building Azure chatbot using Bot Framework and testing it using bot emulator.
- Working on prompt engineering for generative AI for data synthesis, images, and video creation /augmentation. Delivered ML classification solution for Student retention problem for higher education
- Designed, built, and implemented classification models like random forest classifier, Artificial Neural network classifier, and Multiclass boosted decision tree. Experienced in optimization of hyperparameters tuning to obtain the best model, the best model is evaluated for metrics. Deploying the best model as a Kubernetes service.

Client: Statefarm, Bloomington, IL

Mar 2020 to Jul 2022

Role: Gen AI & ML Engineer

Responsibilities:

- Designed and developed Artificial Intelligence situational awareness and threat intelligence systems leveraging Artificial Neural Networks (ANNs), Deep Learning, and Expert Systems.
- Artificial Intelligence (AI) Artificial Neural Networks (ANNs) and Expert Systems architecture for threat intelligence analysis and situational awareness (SA) automated communications. Machine learning application development for pre-cursor Artificial Intelligence systems. Quantum computing systems application development with Python on Quantum Machine Instruction (QMI).
- Expert in supporting national defence operations and networks: C2BMC, DOD C&C practices, JWICS/SIPR/NIPR, general intelligence collection/analysis, and warfighting capabilities. Initial engineering of Artificial Intelligence (early stage) systems for cybersecurity hunter-killer operations.
- Guided and designed overall technology strategy for the Special Agent in Charge (SAC). Served as threat intelligence subject matter expert to the Joint Terrorism Taskforce (JTTF) team members and federal officers.
- Worked on PostgreSQL (pgAdmin), Cassandra (DataStax), and SQL Developer tools to query and extract logs from BPM and MTAS databases. Built different use cases and extensively worked on Jupyter Notebook for Data Cleaning, converted data into structured format, removed outliers, dropped irrelevant columns & missing values, imputed missing values with median/mode/average/min/max other statistical methods

- Worked on libraries like NumPy, Pandas, SciKit Learn, math plot, seaborn, psycopg2, etc. Using machine learning techniques supervised, unsupervised, reinforcement learning and understanding the requirement & design for AI/ML use cases
- Experienced in hyperparameter tuning and model optimization for improved performance.
- Proficient in feature engineering techniques to enhance model accuracy.
- Skilled in evaluating machine learning models using metrics such as accuracy, precision, recall, and F1-score.
- Familiar with deploying machine learning models as REST APIs using Azure services.
- Experienced in working with large datasets and ensuring data quality for machine learning projects.
- Proficient in using Jupyter Notebooks for interactive data analysis and model development.
- Strong understanding of time-series data analysis and forecasting techniques.
- Familiar with integrating machine learning models into production environments using Azure DevOps.
- Capable of collaborating with cross-functional teams to deliver data-driven solutions
- Developed and scaled machine learning and deep learning models like Logistic Regression, Random Forest, Gradient Boosting Machines, SVM (Support Vector Machines), etc. for classification
- Programmed using Python to prototype and deploy Machine Learning, Deep Learning, Predictive models, Probabilistic and Statistical Modelling approaches with user interface development
- Build, train, and deploy ML models on AWS with Amazon Sage Maker. Designing and executing processes related to machine learning and predictive modelling, data mining, and research on large-scale complex data sets using statistical models, graph models, text mining, and other modern techniques
- Provide guidance and direction to ISSE/ISSM/ISSO personnel for correct artifact development and risk mitigation strategy formulation for compliance. Advised on all emerging technologies to include Artificial Intelligence/Machine Learning, Data Science/Big Data Analytics, Cloud Architecture, and Mobility. Worked with time series data, and performed various statistical and (AI) artificial intelligence, (ML) machine learning algorithms such as LMS, regression, filtering, correlation, and neural network. Designed the (AI) artificial intelligence, (ML) Machine learning data pipeline for regular monitoring and performance evaluation of the deployed ML models.
- Built codebase for a natural language processing and (AI) artificial intelligence, (ML) machine learning framework.
- Tested raw data and executed performance scripts. Shared responsibility for the administration of Hadoop, Hive, and Pig. Developed Map Reduce programs to parse the raw data, populate staging tables, and store the refined data in partitioned tables in the EDW.

Environment: Artificial Intelligence, Aws, Gen AI, Hive, ML, Regression, Regression, Cyber Security, Sage maker, PostgreSQL, EDW, JTTF.

Client: World Fuel Services, Miami, FL

Sep 2018 to Feb 2020

Role: Sr.Devops Engineer & SRE Engineer

Responsibilities:

- Worked as DevOps and AWS engineer in Automating, Configuring, and Deploying instances on **AWS** cloud environments. Also Responsible for maintaining and expanding **AWS** infrastructure using **AWS** stack especially worked with database setup and maintenance on **AWS EC2** and migrated applications from customer on-premises datacentre to the cloud.
- Able to create scripts for system administration and **AWS** using languages such as BASH and Python, created Lambda functions to upload code and to check changes in S3, and DynamoDB table.
- Migrated the current Linux environment to the **AWS/RHEL**(RedHat) Linux environment and used the auto scaling feature. Involved in Remediation and patching of Unix/Linux Servers. Created S3 buckets and maintained and utilized the policy management of S3 buckets and Glacier for storage and backup on **AWS**. Developed a notification system using LAMP, PHP, MVC, and Amazon Web services.
- Created monitors, alarms, and notifications for EC2 hosts using **CloudWatch**. Worked on the creation of source code in **Git** and branching, creating labels, and merging of codes on Git for **QA** testing and release.
- Integrating **Jenkins** with **GitHub** for continuous integration and deployment of the code by enabling **Git** hooks, the build jobs are automatically created once the changes are made to the code by the dev team. Worked on using **Maven** as build tools on Java projects for the development of build artifacts on the source code.
- Worked on to setup **Docker** to automate container deployment through **Jenkins** and worked on **docker** container to create **docker** images for different environments.
- Used **Ansible** tower to auto-deploy servers rapidly as per the requirement and also implemented and designed **AWS** virtual servers by Ansible roles to ensure deployment of web applications.
- Maintained high availability clustered and standalone server environments and refined automation components with scripting and configuration management using **Ansible**.
- Worked on building scripts and automated solutions using various scripting languages like **Python**, **Ruby**, and **YAML**. Developed **docker** images using a **docker** file and docker container snapshots and managed **docker** volumes also deployed Docker swarm using Ansible.
- Worked in using **Kubernetes** for container management that runs **docker** containerized applications in a cluster of **EC2** instances in a **Linux** environment. Converted existing **terraform** modules that had version conflicts to utilize cloud formation during **terraform** deployments to enable more control of missing capabilities.

Understand the latest features like (Azure DevOps, OMS, NSG Rules, etc..,) introduced by Microsoft Azure and utilize it for existing business applications

- Creating, validating, and reviewing solutions and effort estimates of converting existing workloads from classic to ARM-based Azure Cloud Environment
- Developed automation system using PowerShell scripts and JSON templates to remediate the Azure services. Worked on implementing backup methodologies by Power Shell Scripts for Azure Services like Azure SQL Database, Key Vault, Storage blobs, App Services, etc.
- Created Azure services using ARM templates (JSON) and ensured no changes in the present infrastructure while doing incremental deployment. Set up GCP Firewall rules to allow or deny traffic to and from the VM's instances based on specified configuration and used GCP cloud CDN (content delivery network) to deliver content from GCP cache locations drastically improving user experience and latency. Extensively worked on various GCP infrastructure design and implementation strategies and experienced in designing, architecting, and implementing scalable cloud-based web applications using AWS and GCP.
- Configuring and deploying instances on GCP environments and Data Centers and building/maintaining Docker container clusters managed by Kubernetes, Bash, Git, and Docker on GCP (Google Cloud Platform).

Environment: AWS, Git, GitHub, Jenkins, Nexus, MQ server, Maven, Ansible, Shell, Python, Ruby, YAML, Docker, Kubernetes, cloudwatch, Terraform.

Client: Centene, St.Louis,MN

May 2017 to Aug 2018

Role: AwsDevOps Engineer

Responsibilities:

- Developed **AWS Cloud Formation** templates to create **VPC, Subnets, Routetables, EC2 instances, ELB, Security Groups, RDS, DynomoDB, CloudWatch, Elastic search clusters, Elastic cache clusters**
- Writing **Ansible** playbooks and Modules to automate our **Build/deployment** process and do an overall process improvement to any manual processes.
- Creating user and group accounts and allowing access using **IAM** setting up own **VPC** networks and security groups and **Peering VPC** to Other VPCs.
- Worked on **Docker** components like **Docker Engine , Hub, Compose, and Registry** .Worked on **EC2** Container Service plug-in in **JENKINS** which automates the Jenkins **master-slave** configuration by creating temporary slaves. Working on **Docker for** building and pushing the images in various environments
- Used **Ansible** Playbooks to set up **Continuous Delivery** Pipeline and developed Python modules for Ansible customization. Deployed **microservices** , including provisioning

AWS environments using **Ansible** Playbooks and Modules. Working with **EC2** Container Service and integrating with **JENKINS**

- Set up Docker to automate container deployment through **Jenkins** and Worked on Docker container to create Docker images for different environments.
- Troubleshoot the build issue during the **Jenkins** build process. Scheduled **Jenkins** jobs by integrating **BITBUCKET**, **Maven**, and **Nexus** for the automated builds using **Groovy**, and **Shellscripts**.
- Worked Extensively in Docker to assist with **auto-scaling**, **continuous integration**, and rolling updates with **no downtime**.
- Used Confluence for technical documentation, and setting up a knowledge base. Performed application server builds in the **EC2** environment and monitored them using **Cloud Watch** and **Dynatrace**.
- Written **Docker Files** according to the Requirements.
- Created and Maintained **Docker files** in Source Code Repository build images and ran containers for microservices applications like ECS and **Openshift**.
- Deployment of binary code in the form of JAR, WAR, and EAR files to **JBoss** and **Apache Tomcat** server from the **S3** Bucket.
- Worked extensively on Adminstrating troubleshooting, configuring, and deploying enterprise applications in the **JBoss web server** and **Apache-Tomcat**. Worked closely with Project Managers, Developers, and QA Analysts to ensure successful software build and deployment for test, staging, and production environment. Acted as build and release engineer, deployed the services by VSTS (Azure DevOps) pipeline. Created and Maintained pipelines to manage the IAC for all the applications
- Assigned RBAC policies Confidential group level and user level as per the LTA created for the services implemented new (Automation account, scheduler, notification hub, IOT Hubs, Batch, and other services). Wrote power shell scripts to create the parameter files automatically for all the services in Azure Resource Manager
- Create a Virtual Network on Windows Azure to connect all the servers. Handle escalated Support tickets till closure for MS Azure PaaS platform. Configured AD connect to configure federation with on-premises ADFS and Azure AD. Experience is using Microsoft Azure. Deployed hybrid cloud model with Anthos to shift the workload between Anthos and cloud GCP
- Integrated the f5 client VPN with GCP Anthos to create the secured tunnel between the cloud to on-premise. Deployed the resources to Google Anthos by using the terraform code and terraform backend in GCP buckets.
- Building/Maintaining Docker/ Kubernetes container clusters managed by Kubernetes Linux, Bash, GIT, Docker, on GCP

Environment: RHEL7.X, Amazon Linux, Ubuntu, AWS (EC2, ECS, ECR, EFS, VPC, IAM, S3, cloud formation, cloud watch and Route53), GIT, GITHUB, Terraform, Maven, Nexus Jenkins, Docker, Ansible, Dynatrace, JBoss, Apache Tomcat, Python, groovy, Ruby, Bash scripting

Client: Master Card, St.Louis,MO

Feb 2016 to Apr 2017

Role: Build and Release & DevOps Engineer

Responsibilities:

- Creating and maintaining scalable and secure infrastructure using EC2 with elastic load balancing and Auto scaling groups in the **AWS** cloud. Creating user and group accounts allowing access using **IAM** and setting up own **VPC** networks and security groups.
- Configuring the **VPC inbound** and outbound networks using public and private **Subnets** according to the requirements. Cloud infrastructure maintenance using EC2, S3, Cloud watch, Cloud Formation, Route 53.
- Responsible for writing custom **chef** cookbooks to install packages and running services over the environment having a number of servers.
- Responsible for the creation of **chef recipe** files to install tomcat instances and to manage configuration files for multiple applications. Written 12 different cookbooks using ruby containing multiple recipes to achieve automation for middleware installation, Environment readiness, and deployment activities. Rapid-provisioning and Configuration Management for Linux, and Windows using **Chef** and Cloud Formation Templates on Amazon Web Services
- Have implemented chef provisioning for the creation of on-demand instances and instance infrastructure using the chef-client and bootstrapping using plugins against AWS EC2. Configured and integrated the servers for different environments to automatically provision and configuration management of Linux instances using the CM/ Provisioning tool Chef.
- Working on **Docker for** building and pushing the images in various environments. Using **Docker** in the process of implementing high-level API which provides lightweight containers to run the applications.
- Working with source code management tool **GIT** to maintain the versions of the committed code. Using MAVEN as a build tool on **Java** projects for the development of build artifacts on the source code.
- Extensively working on **CI** and **CD** processes by automating build on a time basis using **Jenkins**. Responsible for the **management**, User Management, and End-End build and deploy setup using Jenkins.
- Scheduled **Jenkins** jobs by integrating GitHub, Maven, and Nexus for the automated builds using **Shellscripts**. Worked as a **system administrator** for the build and deployment process on the enterprise server.
- Using **Nagios** for network and server monitoring.

Environment: RHEL 6.x,7.1, AWS (EC2, VPC, IAM, S3, cloud formation, cloud watch, and Route53), GIT, GitHub, Maven, Jenkins, Docker, Chef, Nagios, chef, JBoss, Apache Tomcat, Ruby, Bash scripting